INCH-POUND

MIL-PRF-15733/59C
4 August 2020
SUPERSEDING
MIL-PRF-15733/59B
2 June 2009

PERFORMANCE SPECIFICATION SHEET

# FILTERS, RADIO FREQUENCY INTERFERENCE, HERMETICALLY SEALED, STYLE FL64

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

Part or Identifying Numbers (PIN's) M15733/59-0001 through M15733/59-0006 are inactive for new design after 2 June 2009. For new design see table V for supersession data.

The requirements for acquiring the filters described herein shall consist of this specification sheet and MIL-PRF-15733.

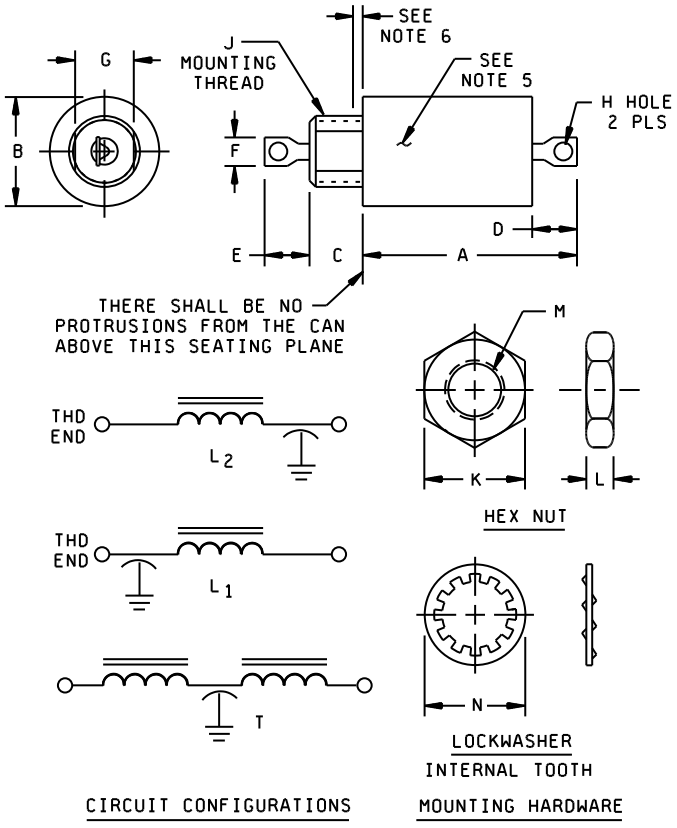



FIGURE 1. Filter and hardware dimensions, and circuit configurations.

AMSC N/A

FSC 5915

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| Dash no. | A | | B | | C | | D | | E | | F | G | | H | | J |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | Typ | Min | Max | Min | Max | |
| 0001 | --- | .743 (18.87) | --- | .416 (10.57) | .175 (4.45) | .205 (5.21) | .100 (2.54) | .190 (4.83) | .100 (2.54) | .190 (4.83) | .119 (3.02) | .190 (4.83) | .210 (5.330 | .045 (1.14) | .075 (1.91) | .250-28 UNF-2A |
| 0002 | --- | .743 (18.87) | --- | .416 (10.57) | .175 (4.45) | .205 (5.21) | .100 (2.54) | .190 (4.83) | .100 (2.54) | .190 (4.83) | .119 (3.02) | .190 (4.83) | .210 (5.330 | .045 (1.14) | .075 (1.91) | .250-28 UNF-2A |
| 0003 | --- | .743 (18.87) | --- | .416 (10.57) | .175 (4.45) | .205 (5.21) | .100 (2.54) | .190 (4.83) | .100 (2.54) | .190 (4.83) | .119 (3.02) | .190 (4.83) | .210 (5.330 | .045 (1.14) | .075 (1.91) | .250-28 UNF-2A |
| 0004 | --- | .743 (18.87) | --- | .416 (10.57) | .175 (4.45) | .205 (5.21) | .100 (2.54) | .190 (4.83) | .100 (2.54) | .190 (4.83) | .119 (3.02) | .190 (4.83) | .210 (5.330 | .045 (1.14) | .075 (1.91) | .250-28 UNF-2A |
| 0005 | --- | .743 (18.87) | --- | .416 (10.57) | .175 (4.45) | .205 (5.21) | .100 (2.54) | .190 (4.83) | .100 (2.54) | .190 (4.83) | .119 (3.02) | .190 (4.83) | .210 (5.330 | .045 (1.14) | .075 (1.91) | .250-28 UNF-2A |
| 0006 | --- | .743 (18.87) | --- | .416 (10.57) | .175 (4.45) | .205 (5.21) | .100 (2.54) | .190 (4.83) | .100 (2.54) | .190 (4.83) | .119 (3.02) | .190 (4.83) | .210 (5.330 | .045 (1.14) | .075 (1.91) | .250-28 UNF-2A |
| 0007 | --- | 1.170 (29.72) | --- | .416 (10.57) | .175 (4.45) | .205 (5.21) | .100 (2.54) | .190 (4.83) | .100 (2.54) | .190 (4.83) | .119 (3.02) | .190 (4.83) | .210 (5.330 | .045 (1.14) | .075 (1.91) | .250-28 UNF-2A |
| 0008 | --- | 1.170 (29.72) | --- | .416 (10.57) | .175 (4.45) | .205 (5.21) | .100 (2.54) | .190 (4.83) | .100 (2.54) | .190 (4.83) | .119 (3.02) | .190 (4.83) | .210 (5.330 | .045 (1.14) | .075 (1.91) | .250-28 UNF-2A |

FIGURE 1. Filter and hardware dimensions, and circuit configurations - Continued.



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| Dash no | Hex nut | | | | | Lockwasher | |
|---|---|---|---|---|---|---|---|
| | K | | L | | M | N | |
| | Min | Max | Min | Max | Thread | Min | Max |
| 0001 | .307 (7.80) | .317 (8.05) | .088 (2.24) | .098 (2.49) | .250-28 UNF-2B | .395 (10.03) | .410 (10.41) |
| 0002 | .307 (7.80) | .317 (8.05) | .088 (2.24) | .098 (2.49) | .250-28 UNF-2B | .395 (10.03) | .410 (10.41) |
| 0003 | .307 (7.80) | .317 (8.05) | .088 (2.24) | .098 (2.49) | .250-28 UNF-2B | .395 (10.03) | .410 (10.41) |
| 0004 | .307 (7.80) | .317 (8.05) | .088 (2.24) | .098 (2.49) | .250-28 UNF-2B | .395 (10.03) | .410 (10.41) |
| 0005 | .307 (7.80) | .317 (8.05) | .088 (2.24) | .098 (2.49) | .250-28 UNF-2B | .395 (10.03) | .410 (10.41) |
| 0006 | .307 (7.80) | .317 (8.05) | .088 (2.24) | .098 (2.49) | .250-28 UNF-2B | .395 (10.03) | .410 (10.41) |
| 0007 | .307 (7.80) | .317 (8.05) | .088 (2.24) | .098 (2.49) | .250-28 UNF-2B | .395 (10.03) | .410 (10.41) |
| 0008 | .307 (7.80) | .317 (8.05) | .088 (2.24) | .098 (2.49) | .250-28 UNF-2B | .395 (10.03) | .410 (10.41) |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses.
3. Metric equivalents are given for general information only.
4. All filters shall be supplied with mounting hardware.
5. Terminal identification (nonsymmetrical filters): The case shall be marked at the threaded end of the filter with the symbol “C” or the symbol “L”, as follows:

| Circuit | Symbol |
|---|---|
| $L_1$ | C |
| $L_2$ | L |

6. Imperfect thread or undercut optional; .047 (1.19) max, 2 places.
7. For use in applications where mounting panel thickness does not exceed .062 inch (1.57 mm).
8. Recommended mounting torque: 48 ounce-inch, maximum.

FIGURE 1. Filter and hardware dimensions, and circuit configurations - Continued.



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TABLE I. Electrical Characteristics.

| Dash no. | Circuit configuration | Maximum rated current (amp) | Maximum rated voltage | | | | Voltage drop (volts DC) or (volts rms) max | Minimum full load insertion loss (dB) in accordance with MIL-STD-220 over full temperature range -55°C to +125°C | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Volts, DC | | AC Vrms 60 Hz or 400 Hz | | | | | | | |
| | | | +85°C to -55°C | +125°C | +85°C to -55°C | +125°C | | 150 kHz | 300 kHz | 1.0 MHz | 10.0 MHz | 1.0 GHz |
| 0001 1/ | $L_2$ | 1.0 | 200 | 200 | 150 | 150 | .250 | 12 | 23 | 44 | 70 | 70 |
| 0002 1/ | $L_1$ | 1.0 | 200 | 200 | 150 | 150 | .250 | 12 | 23 | 44 | 70 | 70 |
| 0003 1/ | $L_2$ | 3.0 | 200 | 200 | 150 | 150 | .150 | 7 | 13 | 29 | 67 | 70 |
| 0004 1/ | $L_1$ | 3.0 | 200 | 200 | 150 | 150 | .150 | 7 | 13 | 29 | 67 | 70 |
| 0005 1/ | $L_2$ | 5.0 | 200 | 200 | 150 | 150 | .075 | 7 | 12 | 24 | 57 | 70 |
| 0006 1/ | $L_1$ | 5.0 | 200 | 200 | 150 | 150 | .075 | 7 | 12 | 24 | 57 | 70 |
| 0007 | T | 2.0 | 200 | 200 | 150 | 150 | .200 | 11 | 19 | 42 | 70 | 70 |
| 0008 | T | 4.0 | 200 | 200 | 150 | 150 | .250 | 10 | 17 | 32 | 70 | 70 |

1/ Inactive for new design.





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REQUIREMENTS:

Dimensions and configuration: As specified in figure I.

Case: Metal.

Case and hardware finish: In accordance with MIL-PRF-15733. Pure tin is prohibited as an undercoat and as a final finish.

Seal: Glass to metal.

Terminals: Solderable (see figure 1). Pure tin is prohibited as an undercoat and as a final finish.

Operating temperature range: -55°C to +125°C.

Rated voltage: See table I.

Rated current: See table I.

Voltage drop: See table I.

Insertion loss: See table I.

Seal: In accordance with MIL-PRF-15733 and MIL-STD-202-112; test condition A or D.

Temperature rise: In accordance with MIL-PRF-15733. Temperature rise shall be +25°C, maximum.

Dielectric withstanding voltage: In accordance with MIL-PRF-15733. The following exception shall apply:

Test voltage: 2 times rated voltage (V dc).

Barometric pressure (reduced): In accordance with MIL-PRF-15733 and MIL-STD-202-105; test condition D.

Inrush transients test: the filters shall be subjected to test by instantaneous application of 115 V rms, 400 Hz to the filter with the output unloaded. The source impedance shall be less than 0.5 ohm and its energy capability shall not limit the inrush current to the filter. The filters shall be subjected to this inrush a minimum of 10 times, with a maximum time of 5 minutes between applications. The input shall always be into the inductor for non-symmetrical filters.

Measurements after test: Insulation resistance (at +125°C) and insertion-loss shall meet initial requirements.

Transient voltage test: The filters shall be subjected to the overvoltage allowed by MIL-STD-704 for ac systems. The following details and exceptions shall apply:

Preconditioning: Filter shall be energized at 115 V rms, 400 Hz carrying rated current for 15 minutes minimum.

Transient application: The output load shall be removed and the input voltage shall be stepped form 115 V rms to 180 volts within 20 milliseconds of removal of the load. The step from 115 to 180 volts shall be accomplished with a rise time of less than 60 milliseconds. The 180 volts shall be maintained for 100 milliseconds minimum. This cycle shall be repeated 10 times.

Measurements after test: Insulation resistance (at +125°C) and insertion-loss shall meet initial requirements.



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Insulation resistance: In accordance with MIL-PRF-15733. The following details and exceptions shall apply:

Test potential: Rated dc voltage.

Insulation resistance: Shall not be less than 100 megohms at +25°C and not less than 5 megohms at +125°C, when measured between the case (ground) and all terminals connected together.

Voltage drop: In accordance with MIL-PRF-15733. Maximum voltage drop at +25°C shall be as specified in table I.

Insertion loss: In accordance with MIL-PRF-15733 and table I at +25°C and temperature extremes.

Terminal strength: In accordance with MIL-PRF-15733 and MIL-STD-202-211; test condition A (pull).

Applied force: 4.5 pounds.

Salt atmosphere (corrosion): In accordance with MIL-PRF-15733 and MIL-STD-202-101; test condition B.

Shock (specified pulse): In accordance with MIL-PRF-15733 and MIL-STD-202-213; test condition I.

Vibration, high frequency: In accordance with MIL-PRF-15733 and MIL-STD-202-204; test condition D (20 G).

Moisture resistance: In accordance with MIL-PRF-15733. The following exceptions shall apply:

Measurements after 24-hour drying period:

Dielectric withstanding voltage: Voltage at 1.8 times rated dc voltage.

Insulation resistance and insertion-loss shall meet initial requirements.

Life: In accordance with MIL-PRF-15733 and MIL-STD-202-108. The following details and exceptions shall apply:

Test condition D (1,000 hours) for qualification inspection, test condition B (250 hours) for group C inspection.

Measurements after test at +25°C:

Dielectric withstanding voltage at 1.8 times rated dc voltage.

Insulation resistance and insertion-loss shall meet initial requirements.

Voltage conditioning: Voltage conditioning shall consist of applying rated ac voltage 400 Hz to the part at +125°C. The voltage shall be applied for a minimum of 250 hours. Charging and discharging current shall be limited within the range of 30 to 150 milliamperes. The test circuit shall be in accordance with figure 2. $E_1$ shall be continuously monitored and if it falls below 95 percent of $E_T$, the test shall be stopped and the defective part removed. Time counted toward total test time shall be the time that $E_1$ is greater than 95 percent of $E_T$. After the completion of 250 hours and while remaining at +125°C, the insulation resistance shall be measured and shall be 5 megohms minimum.



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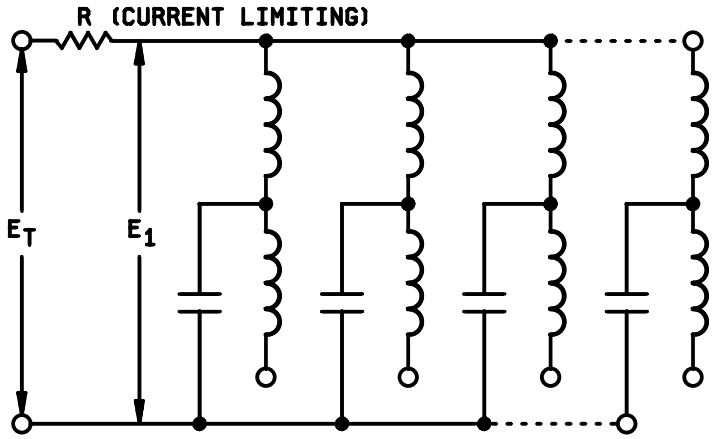



FIGURE 2. Voltage conditioning test circuit.

VERIFICATION

Qualification inspection: Requirements of this specification sheet not currently required in MIL-PRF-15733 shall be included in qualification inspection as group VI (see table II).

TABLE II. Qualification inspection. 1/

| Examination or test | Number of samples to be inspected | Number of defectives allowed |
|---|---|---|
| Group V | | |
| Transient voltage | 4 | 0 |
| Inrush transients | 4 | |

1/ This table is an addition to table III (qualification inspection) of MIL-PRF-15733.

Conformance inspection: Conformance inspection shall be as specified in MIL-PRF-15733 except group A inspection shall be as specified in table III of this specification sheet, in the order shown. Lots that have greater than 5 percent rejects shall be considered reject lots. Reject lots may, at the option of the supplier, be resubmitted to 100 percent group A inspection; however, resubmitted lots which have greater than 2 percent rejects shall be rejected and may not be resubmitted for reinspection.

Group C inspection shall be as specified in MIL-PRF-15733 and shall include the tests specified in table IV of this specification sheet.



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TABLE III. Group A inspection. 1/

| Test or inspection | Sampling procedure |
|---|---|
| Thermal shock | 100% |
| Voltage conditioning | 100% |
| Dielectric withstanding voltage | 100% |
| Insulation resistance | 100% |
| Voltage drop | 100% |
| Insertion loss | 100% |
| Seal | 100% |
| Visual and mechanical inspection | See MIL-PRF-15733. Group A inspection. |

1/ This table replaces table IV (Group A inspection) of MIL-PRF-15733.

TABLE IV. Group C inspection. 1/

| Examination or test | Number of samples to be inspected | Number of defectives allowed |
|---|---|---|
| Subgroup 6 | | |
| Transient voltages | 4 | 0 |
| Inrush transients | 4 | |

1/ This table is an addition to table VI (Group C inspection) of MIL-PRF-15733.

Part or Identifying number (PIN): M15733/59- (dash number from table I).

Supersession data for new design: See Table V

TABLE V. Supersession data

| Inactive PIN | For new design use | |
|---|---|---|
| | Specification sheet | PIN |
| M15733/59-0001 | MIL-PRF-28861/5 | M28861/05-005TB |
| M15733/59-0002 | MIL-PRF-28861/5 | M28861/05-004TB |
| M15733/59-0003 | MIL-PRF-28861/5 | M28861/05-008TB |
| M15733/59-0004 | MIL-PRF-28861/5 | M28861/05-007TB |
| M15733/59-0005 | MIL-PRF-28861/5 | M28861/05-011TB |
| M15733/59-0006 | MIL-PRF-28861/5 | M28861/05-010TB |



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Referenced documents. In addition to MIL-PRF-15733, this specification sheet references the following documents:

MIL-PRF-28861/5
MIL-STD-202-101
MIL-STD-202-105
MIL-STD-202-108
MIL-STD-202-112
MIL-STD-202-204
MIL-STD-202-211
MIL-STD-202-213
MIL-STD-220
MIL-STD-704

Changes from previous issue. The margins of this specification sheet are marked with vertical line to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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